CLINOMA PLATFORM

FIRST PAPER CAMP

Questions & Exercises Booklet

Day 2: Pediatrics & Hematology/Oncology

Prepared by: Clinoma Platform Team

CHAPTER 01

Model 1

CLINOMA PLATFORM · PREMIUM STUDY CAMP

Part I: Short Essay & Enumerate Questions

Q1: Enumerate four clinical manifestations of "Neglected" (untreated) Hypothyroidism during infancy and childhood (by 3-6 months of age).

Q2: Enumerate four primary (congenital) causes of short stature.

Q3: Enumerate four acute or long-term micro-vascular complications of Type 1 Diabetes Mellitus.

Part II: Short Answered Questions

Q4: Define "Idiopathic Short Stature (ISS)" according to the curriculum.

Q5: Mention the criteria used for the biochemical diagnosis of Diabetes Mellitus.

Q6: List four environmental risk factors or hypotheses that act as initiators or accelerators for beta-cell destruction in Type 1 Diabetes.

Part III: Problem Solving (Clinical Cases)

Case 1

A pediatric team is performing a routine neonatal screening for a 5-day-old baby via a heel-prick blood test. The laboratory results return showing a significantly elevated Thyroid Stimulating Hormone (TSH) level of 35 uIU/ml and a low Thyroxin (T4) level.

A): What is the provisional diagnosis for this newborn?

B): Why is neonatal screening highly important for this specific condition even if the baby looks completely normal at birth?

Case 2

A 7-year-old girl is brought by her parents due to a noticeable downward shift in her growth rate that originally started when she was about 6 months old. Her current growth curve remains below, but completely parallel to, the 3rd percentile for height. Her bone age radiograph is delayed but matches her height age, and she has no chronic illnesses or endocrine abnormalities. Her parents note a family history of "late bloomers."

A): What is the most likely diagnosis among the normal variants of short stature?

B): What is the expected long-term adult height outcome for this child?

CHAPTER 02

Model 2

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Part I: Short Essay & Enumerate Questions

Q1: Enumerate four general risk factors (other than diet and lack of exercise) that increase the risk of a child becoming overweight.

Q2: Enumerate four clinical features of Type 1 Diabetes Mellitus (Classic presentation and other symptoms).

Q3: Enumerate four components of the detailed history required to evaluate a child presenting with Short Stature.

Part II: Short Answered Questions

Q4: Define "Childhood Obesity" and "Overweight" using BMI percentile charts for children over 2 years of age.

Q5: Mention the three components of the "Triple Cascade" (Pathophysiology) that occur in a patient during severe insulinopenia leading to DKA.

Q6: List four clinical symptoms or signs that appear within weeks or months after birth in severe cases of Congenital Hypothyroidism.

Part III: Problem Solving (Clinical Cases)

Case 1

A 10-year-old boy is brought to the clinic for weight evaluation. His Body Mass Index (BMI) is calculated and found to be on the 98th percentile for his age and sex. On physical examination, his linear growth (height velocity) is found to be completely normal and consistent with his target height.

A): Based on his BMI percentile, how is this child's weight condition classified?

B): Does his normal/consistent linear growth suggest Nutritional Obesity or an underlying Endocrinopathy?

Case 2

A 5-year-old girl is brought by her parents due to severe short stature. On examination, her height is 2.5 Standard Deviations (SD) below the mean for her age. All her body parts are in the usual proportions (proportionate short stature). Laboratory tests reveal a high TSH level and a very low Free T4 level.

A): What is the underlying cause of this child's short stature?

B): What is the treatment of choice for this condition?

CHAPTER 03

Model 4

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Part I: Short Essay & Enumerate Questions

Q1: Enumerate four general protective measures or clinical practices that must be STRICTLY AVOIDED in a child diagnosed with Hemophilia.

Q2: Enumerate four clinical manifestations or multi-organ complications caused by severe systemic Iron Overload (Hemosiderosis) in Thalassemia Major patients.

Q3: Enumerate four clinical manifestations resulting directly from Leukopenia (Neutropenia) and Thrombocytopenia in a patient with Aplastic Anemia.

Part II: Short Answered Questions

Q4: Define "Erythropoiesis" and mention the key hormone that drives it along with its primary organ of production.

Q5: Mention four specific oxidant triggers (medications or acute conditions) that can precipitate severe intravascular hemolysis in a child with G6PD deficiency.

Q6: Define "Koilonychia" and state the specific type of pediatric nutritional anemia it is classically associated with.

Part III: Problem Solving (Clinical Cases)

Case 1

A 4-year-old boy is brought to the emergency room with a painful swelling, warmth, and limited range of motion in his left knee joint following a minor slip while playing. His mother reports a positive family history of a chronic bleeding disorder on the maternal side affecting her brother. Laboratory investigations show: a completely normal platelet count, normal Bleeding Time (BT), normal Prothrombin Time (PT), but a significantly prolonged activated Partial Thromboplastin Time (aPTT).

A): What is the most likely provisional diagnosis?

B): What is the standard supportive first-aid protocol used for managing this child's joint bleed (hemarthrosis) alongside specific factor replacement?

Case 2

A 6-year-old girl is evaluated for chronic, intermittent episodes of mild jaundice, fatigue, and a palpable left upper quadrant mass (splenomegaly). Her father notes that he had to undergo a splenectomy during his adolescence for a similar condition. A peripheral blood smear is performed, revealing numerous small, densely stained red blood cells that completely lack a central pale area (spherocytes). The Direct Antiglobulin Test (Coombs test) is negative.

A): What is the most likely hereditary diagnosis?

B): Name the specific laboratory test considered the absolute diagnostic test of choice to confirm this red blood cell membrane defect.